Human Activity Recognition from Wi-Fi CSI Doppler Spectrograms: A CNN Baseline and a Contrastive Extension for Cross-Domain Generalization

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***Abstract*—We study device-free human activity recognition (HAR) from the Doppler spectrograms of Wi-Fi channel-state information (CSI), using the publicly released traces of the SHARP dataset [1], [2]. Starting from the Inception-style convolutional baseline of the course material—trained *only* on measurement set $S1$ and evaluated on the six unseen sets $S2$–$S7$, where it reaches 48.2%—we add and measure, one change at a time, four strategies for closing the cross-domain gap without the hand-engineered front-end of the reference system. (i) *Supervised contrastive pre-training*: a batch-normalized encoder trained with a SupCon objective and domain-randomization augmentations, with a linear probe on the frozen representation, gives 53.1% and improves *every* measurement set. (ii) *A slower optimizer schedule* reaches 51.3% while fitting the training environment *worse*, showing that on a single-environment training set the two objectives conflict. (iii) *An overlapping sliding window* at the stride used by the dataset authors (30 samples, 91% overlap, $\approx 10\times$ more windows), made affordable by a two-pass, memory-bounded loader, lifts the same baseline to 59.3%. (iv) *Multi-antenna soft decision fusion*: a recording-level group identifier threaded through the data pipeline lets the four per-antenna probability vectors of one physical time window be averaged, adding $+12.8$ points with no retraining and no extra parameters. Together the last two reach $\mathbf{72.1}\%$ ($+23.9$ over the initial baseline) and beat the single-component ablation of the reference system on every unseen set. Notably, contrastive pre-training helps in the data-poor regime but not once the sliding window supplies ten times more windows. The residual gap to the full pipeline ($> 95\%$) is attributable to its strongest-path phase sanitization, which we do not implement.**

I. Introduction

Contactless sensing of human activity is valuable in healthcare, security and ambient monitoring. Commodity Wi-Fi devices can be repurposed as radar-like sensors: as a person moves, their body induces micro-Doppler shifts captured in the channel-state information (CSI) estimated by the receiver, *without recording any video*. After processing, the CSI becomes a *Doppler spectrogram*—a time $\times$ velocity image—whose appearance changes markedly with the activity, so recognition can be cast as image classification with a convolutional neural network (CNN) [1].

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Course project for Neural Networks and Deep Learning. Baseline pipeline by M. Carraro; Supervised Contrastive Learning extension by F. Vezzani.

The central difficulty is *generalization*. Different people move differently, and different rooms—even the same room on different days—distort the signal differently, so a model that classifies perfectly in its training environment may collapse on a new one. The project proposal raises exactly this challenge and suggests a *contrastive / self-supervised* approach to obtain subject- and environment-independent features. Prior systems close the gap mostly through signal processing—robust phase sanitization and multi-antenna fusion—at the cost of a heavy, hand-engineered pipeline [1].

We tackle five-class Wi-Fi HAR under a strict cross-domain protocol (train on one environment, test on six unseen ones) and ask how much of the gap can be closed *without* that front-end: by *learning* invariance with Supervised Contrastive Learning (SupCon) [3] instead of engineering it, by re-using the receiver diversity the dataset already provides, and by feeding the network more of the signal it already has. Every change is implemented and measured *one at a time*, on the same baseline and protocol, so each accuracy figure is attributable to one design decision (Tab. 1). The contributions are:

- **Baseline (M. Carraro).** A faithful plain-PyTorch reimplementation of the dataset pipeline and of the baseline Inception-style CNN, with a cross-domain protocol (train on $S1$, test on $S2$–$S7$) and a per-measurement-set error breakdown.
- **Contrastive representation (F. Vezzani).** A batch-normalized encoder pre-trained with a supervised-contrastive loss and domain-randomization augmentations, evaluated with a linear probe on the frozen features. It improves every set when training data is scarce—and, as we show, stops helping once it is not.
- **Multi-antenna decision fusion.** A group identifier that survives the whole data pipeline recovers which windows are the *same* physical time window seen by the four antennas, so their predictions can be fused by averaging class probabilities: $+10$ to $+13$ points in both data regimes, with no retraining.
- **Front-end and optimization engineering.** A sliding window at the stride used by the SHARP authors (30 samples, 91% overlap) replaces the naïve segmentation—

$+11.1$ points on its own, affordable only through a two-pass, memory-bounded dataset builder; and a controlled comparison of learning-rate and early-stopping settings shows that, with a single training environment, fitting the training set better does *not* imply generalizing better.

Combining the last two takes the same architecture from 48.2% to $\mathbf{72.1}\%$. We use the full SHARP system [1] as the reference upper bound and discuss, quantitatively, which part of the gap each strategy removes and which part is left.

Section II reviews the related work; Section III gives a high-level view of the pipeline and of the changes we evaluate; Section IV details signals and features and Section V the learning frameworks. Results and the comparison against SHARP are in Section VI, concluding remarks in Section VII.

II. Related Work

Wi-Fi sensing and the SHARP system. Device-free HAR from commodity Wi-Fi has matured rapidly thanks to CSI extraction tools and public datasets. The most relevant work for us is SHARP [1], which recognizes activities from the Doppler information in IEEE 802.11ac CSI and targets *environment and person independence* through two ingredients: (i) a robust phase-sanitization step that isolates the multipath components and adopts the *strongest path* as phase reference, and (ii) a *decision-fusion* scheme that runs a compact Inception-style CNN on each of the four receiver antennas and combines their decisions. On the seven measurement sets of the accompanying dataset [2] the full system exceeds 95% even on rooms, days and persons never seen at training time. Those results rest on a heavy, hand-engineered front-end: the paper's own ablation shows that replacing the strongest-path sanitization with the standard reference-antenna method makes the *same* CNN collapse to 39–60% on the unseen sets, i.e. much of the robustness is carried by the pre-processing rather than by the learned representation. Our work keeps the minimal front-end (no strongest-path sanitization) and asks how much cross-domain robustness can be recovered elsewhere: on the *representation* side (contrastive pre-training) and on the *decision* side, by re-implementing SHARP's antenna fusion—which, unlike the sanitization step, costs nothing at training time and turns out to be worth more than any other change we made.

Contrastive representation learning. Contrastive methods learn representations by pulling together "views" that should be similar and pushing apart those that should differ. SimCLR [4] established the self-supervised recipe: two augmentations of an image form a positive pair, all other images in the batch are negatives, and an encoder is trained with a temperature-scaled InfoNCE loss on L_2-normalized embeddings. Supervised Contrastive Learning (SupCon) [3] extends it to the labelled setting—*all* samples sharing the anchor's label become positives—producing tighter, more class-consistent clusters than cross-entropy. We adopt SupCon because its notion of "positive" maps onto our goal: two windows of the same activity should be neighbours *whatever environment, person or antenna they come from*. Combining it with explicit domain-randomization augmentations for cross-domain Wi-Fi HAR on the SHARP data is the extension we contribute, and Section VI quantifies both its gains and the regime in which they vanish.

III. Processing Pipeline

All of our models cast Wi-Fi HAR as the classification of Doppler-spectrogram images and share the *same* data front-end; they differ in *how* the representation is learned and in *how* the per-window decisions are combined. The raw CSI Doppler traces are cut into fixed-size windows of shape $1 \times 340 \times 100$, one antenna at a time (Section IV); from there the pipeline forks:

- the **baseline chain** (M. Carraro) feeds each window to an Inception-style CNN trained end-to-end with cross-entropy, and reads the activity off the softmax output;
- the **contrastive chain** (F. Vezzani) first applies domain-randomization augmentations to build two views of each window, pre-trains a deeper batch-normalized encoder with a Supervised Contrastive loss, then *freezes* the encoder and fits a linear probe on the resulting features;
- the **fusion stage** sits *after* either classifier and needs no retraining: the per-antenna class probabilities of the same physical time window are averaged into one decision.

All chains are trained on a single environment ($S1$) and evaluated under the same cross-domain protocol (unseen $S2$–$S7$), so they can be compared on equal footing against the reference SHARP system [1]; Section V details them.

A. The changes we evaluate

Four modifications were implemented on that skeleton, each isolated so that its effect is attributable (Tab. 1): a *windowing* strategy, a *group-aware* pipeline enabling *decision fusion*, a *contrastive* representation and an *optimizer schedule*. Only the first touches the data loader and only the third trains a new network; fusion is post-processing on an unchanged model, which is what makes its gain remarkable. The two largest—the sliding window and fusion—compose into our best configuration.

IV. Signals and Features

(Dataset pipeline implemented by M. Carraro.)

A. The SHARP data

We use the pre-extracted Doppler traces released with the SHARP dataset [2] (IEEE 802.11ac, 80 MHz): arrays of shape $(T, 100)$ —T time steps, 100 Doppler-velocity bins—one per receiving antenna (`stream_1`…`stream_4`) of a four-antenna monitor. They map to seven *measurement sets* $S1$–$S7$, each a distinct (environment, monitor position, person, day) combination. Only $S1$ is used for training; the others probe generalization to an unseen day ($S2$), an unseen person ($S3$), blocked line-of-sight ($S4$, $S5$), a different room ($S6$), and new room, day and person at once ($S7$).

TABLE 1: The modifications evaluated, the code each required, and their effect on cross-domain accuracy (train on $S1$, test on $S2$–$S7$, one run each). Each is measured against the reference configuration of the first row, or against the same configuration with a sliding front-end.

Strategy	Implementation	Effect
Reference	non-overlapping windows (stride 340, 13,564 windows); reduced-Inception CNN (128,535 par.); Adam 10^{-3}, patience 7	48.2%
Contrastive	augmentations + BN encoder with global average pooling + SupCon loss; linear probe on the frozen encoder	+4.9 (−1.0 sliding)
Slow schedule	Adam 10^{-4}, patience 5, budget 100; nothing else changes	+3.1
Sliding window	stride 30 (91% overlap, 152,320 windows); two-pass pre-allocating loader, splits built separately; patience raised to 12	+11.1
Antenna fusion	group id emitted by the loader and carried through the split; softmax and per-group probability average at inference, same weights	**+12.8**(+10.2 non-ov.)

B. From traces to windows

We use the five core activities present in every environment—**E** (empty room), **J** (jumping), **L** (sitting still), **R** (running), **W** (walking)—matching the five-class task of [1]. Each trace is cut into windows of $W_{\text{len}} = 340$ time samples (≈ 2 s) advancing by a stride S, plus a channel dimension: inputs of shape $1 \times 340 \times 100$. Each antenna stream is an independent labelled sample, as SHARP does at training time; antenna identity is re-used only at *decision* time (Section V-C).

Windowing strategy. The reference configuration uses $S = W_{\text{len}} = 340$, i.e. *non-overlapping* windows: simplest, but wasteful—a trace of length T yields only $\lfloor T/340 \rfloor$ examples and any pattern straddling a boundary is cut in half. It gives 13,564 windows (E: 3012, J: 2160, L: 3056, R: 2684, W: 2652). The alternative is a sliding window: after prototyping $S = 170$ we adopted $S = 30$, the value hard-coded in the SHARP authors' own dataset script—91% overlap and 152,320 windows from the same recordings. They are strongly correlated, so the number of *independent* examples grows far less than tenfold, and a window and its neighbour can fall on opposite sides of the random train/val split, making the validation loss optimistic and perturbing *when* early stopping fires; the test figures are unaffected, the test environments being disjoint.

Group identifiers. The four streams of a recording have identical length, so window k of each starts at sample kS of the same event. The parser therefore keeps the full recording token (`"J2"`, repetition number included) and the loader emits a group id `env_token_k` per window: windows sharing it are the same physical time window seen by different antennas, which is what makes Section V-C possible. The id travels through the split with the labels, and the test `DataLoader` uses `shuffle=False` to stay aligned with it.

A memory-bounded loader. At $S = 30$ the windows (4.98 GB train/val, 15.74 GB test) no longer fit comfortably in a Colab runtime: the original loader collected per-file arrays in a list and called `np.concatenate`, holding list *and* result in RAM at once—affordable at 1.8 GB, not at 20 GB. The rewritten loader makes two passes: the first counts the windows each trace will produce, the second fills one pre-allocated `float32` array and frees each trace as soon as it is windowed. It also takes the environments to load, so train/val and test arrays come from two independent calls rather than slicing one large array, and the copy left by `train_test_split` is deleted explicitly. Peak memory is roughly halved and the windows are identical.

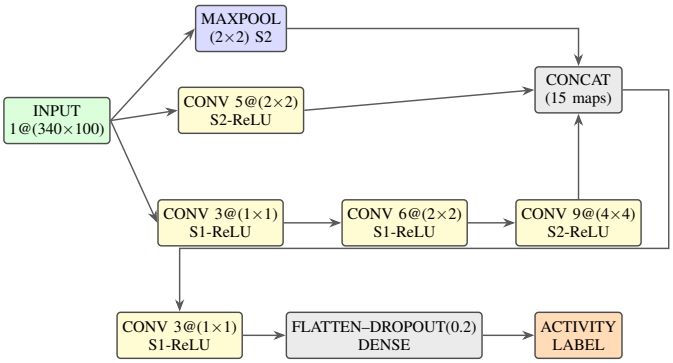


Fig. 1: Baseline single-antenna architecture, a faithful reproduction of the SHARP reduced-Inception module ([1], Fig. 4). `S2` denotes stride 2; every CONV is followed by a ReLU (128,535 parameters).

C. Cross-domain split

Following the SHARP protocol the split is by environment, never random: $S1$ ($S1a, S1b, S1c$) is used for training/validation and every other set is held out; a random split would leak windows from the same person/room/day into both sides and overestimate performance. Within $S1$ we keep 85% for training and 15% for validation (stratified), 2767 / 489 / 10,308 windows for train / val / test at $S = 340$ and 31,103 / 5489 / 115,728 at $S = 30$. Grouping the four antennas collapses the test windows to 2577 and 28,932 distinct physical time windows respectively—the number of decisions the fused evaluation reports.

V. Learning Framework

A. Baseline CNN (M. Carraro)

The baseline reproduces the reduced-Inception module of SHARP ([1], Fig. 4), shown in Fig. 1: three parallel branches (max-pool, strided conv, three stacked convs) turn the $1 \times 340 \times 100$ input into feature maps of identical size 170×50, concatenated on the channel axis ($1{+}5{+}9 = 15$), reduced to 3 channels by a 1×1 convolution, flattened and classified by a dense layer after $\text{Dropout}(0.2)$— **128,535** parameters, exactly the count reported for the SHARP baseline CNN. Training uses cross-entropy, Adam, mini-batches of 64 and *early stopping*, restoring the lowest-validation-loss weights.

Three optimizer schedules were compared with architecture, seed, batch size and split held fixed, because their effect is counter-intuitive. The *fast* schedule (lr $= 10^{-3}$, patience 7, budget 40) stops at epoch 13 (best validation loss 0.993) with the classic overfitting signature: training accuracy climbs to

0.601 while validation accuracy plateaus near 0.52. The *slow* schedule (10^{-4}, patience 5, budget 100) runs for 41 epochs with the two curves almost on top of each other and stops at a training accuracy of only 0.575: the model has not yet fitted its own training set. Reading that as underfitting, the *adopted* schedule keeps 10^{-3} but raises patience to 12, so a plateau no longer ends training prematurely—which matters at stride 30, where each epoch is an order of magnitude slower; on that data it reaches 0.756 training accuracy and stops at epoch 27.

B. Contrastive extension: Supervised Contrastive Learning (F. Vezzani)

1) Motivation and augmentations: Plain cross-entropy on a single environment pushes nothing towards domain-invariant features. Following the *contrastive / self-supervised* direction suggested in the project proposal, we implement *Supervised Contrastive Learning* (SupCon) [3]: an *encoder* is trained so that two windows of the *same activity* are pulled together and windows of different activities pushed apart, *whatever environment, person or antenna they come from*; a linear classifier is then fit on the frozen representation (*linear probe*). Each window needs two augmented "views" that keep the activity intact but perturb nuisance factors—a light domain randomization. All transforms run on-GPU on a whole batch: **amplitude scaling** ($\mathcal{U}(0.8, 1.2)$, rooms and distances reflect the signal with different strength), **additive Gaussian noise** ($\sigma = 0.05$), **circular time shift** (± 40 samples, the activity does not depend on where in the window it starts) and **time/Doppler masking** (SpecAugment-style, a zeroed band of 40 time samples and 15 velocity bins, forcing the encoder to use the whole time–velocity pattern).

2) Encoder and projection head: The encoder $f(\cdot)$ is a deeper, batch-normalized evolution of the baseline's Inception block: the same three-branch front-end (concatenated to 41 channels) plus (i) batch normalization after every convolution, (ii) a 1×1 reduction to 32 channels, (iii) a second downsampling stage ($32 \to 64 \to 128$, output $128 \times 43 \times 13$) and (iv) *global average pooling*, which replaces the baseline's large `flatten`→dense layer with a fixed 128-d representation h that has fewer parameters and is more shift-robust. An MLP *projection head* $g(\cdot)$ ($128 \to 128 \to 128$, ReLU) maps h to an L_2-normalized $\boldsymbol{z} = g(h)$; as prescribed by SupCon the loss is applied to $\boldsymbol{z}$ and the head is *discarded* afterwards, so classification uses h. The whole model has $130{,}952$ parameters against the baseline's 128.5k.

3) SupCon loss: For a batch of L_2-normalized embeddings with labels, each anchor i attracts every other sample sharing its label (its positive set $P(i)$) and repels all others, through a temperature-scaled softmax over cosine similarities:

$$\mathcal{L} = \frac{1}{|I|} \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\boldsymbol{z}_i \cdot \boldsymbol{z}_p / \tau)}{\sum_{a \neq i} \exp(\boldsymbol{z}_i \cdot \boldsymbol{z}_a / \tau)}, \qquad (1)$$

where I is the set of anchors with at least one same-class positive and $\tau = 0.07$. Unlike self-supervised SimCLR [4], using the labels makes *all* same-activity windows in the batch positives—even those from different files or environments—which turns generic contrastive learning into an environment-invariant *activity* representation.

4) Training and evaluation protocol: Everything uses the *same* $S1$-based split as the baseline. In **pre-training** each batch is expanded into two augmented views and encoder and head are trained jointly with Adam (weight decay 10^{-4}) for up to 30 epochs with early stopping on the validation contrastive loss. The **linear probe** then freezes the encoder, pre-computes and standardizes the embeddings and trains a single $\mathrm{Linear}(128, 5)$ layer with cross-entropy—isolating the quality of the representation—while **optional fine-tuning** unfreezes everything at lr 10^{-4}. A t-SNE projection of the test embeddings, colored by activity and by environment, is the invariance diagnostic. The contrastive and probe learning rates were 10^{-3} on the non-overlapping data and 10^{-4} on the sliding-window data, a difference we return to in Section VI.

C. Multi-antenna decision fusion

The dataset ships the four antenna streams *separately*, and both chains so far discard that structure. SHARP keeps the per-antenna training convention but fuses the four decisions at inference [1]; that step is reproduced here as pure post-processing. For a group id g, let A_g be the (up to four) antenna windows carrying it and $\boldsymbol{p}_i = \mathrm{softmax}(f(\boldsymbol{x}_i))$ the class probabilities of window i; the fused decision is

$$\hat{y}_g = \underset{c \in \{1,\ldots,5\}}{\arg\max} \; \frac{1}{|A_g|} \sum_{i \in A_g} p_{i,c}, \qquad (2)$$

i.e. a *soft* vote, preferred to majority voting on the hard labels because a confident antenna can outweigh two unsure ones and because it degrades gracefully when a recording provides fewer than four streams. The reference paper does not detail its own rule, so Eq. (2) is a standard instantiation rather than a byte-for-byte reproduction.

Two details make it correct: the group id must *not* encode the stream index, so the four antennas collide onto one key, and the evaluation `DataLoader` must not shuffle, or the probabilities stop lining up with the stored ids. Fusion also changes the *unit* of evaluation: one decision per physical time window instead of one per antenna.

VI. Results

All numbers come from the stored notebook outputs, one run per configuration, under the same protocol: train on $S1$, test on the unseen $S2$–$S7$. Tab. 2 collects them per set and Fig. 4 plots the four milestones.

A. Baseline

On the pooled unseen test set the baseline reaches $\mathbf{48.2}\%$ (test loss 1.052), from 45.5% ($S2$) to 52.3% ($S7$). Its confusion matrix (Fig. 2 (a)) has two dominant error modes: *sitting still* is absorbed by *empty* ($L \to E$: 0.76), both near-static and concentrated at zero Doppler velocity, and *running/walking* are heavily confused (0.39 and 0.28).

TABLE 2: Per-set accuracy (%), training on $S1$ and testing on the unseen sets, one run per column. Non-overlapping windows: baseline under the fast (10^{-3}) and slow (10^{-4}) schedules, SupCon linear probe, and fusion on the slow-schedule model. Sliding window (stride 30): baseline, SupCon probe and fusion. Fusion always re-uses the weights of the baseline to its left. SHARP columns from [1] (Tables 3 and 4).

	Non-overlapping windows				**Sliding window (stride 30)**			**SHARP [1]**	
Set	Base (fast)	SupCon	Base (slow)	+ Fusion	Base	SupCon	+ Fusion	Full	Abl.
$S2$	45.5	51.1	50.9	62.6	60.8	59.3	77.9	99.8	59.8
$S3$	51.4	59.7	55.9	68.8	61.6	61.5	75.2	99.3	58.0
$S4$	48.5	51.9	50.6	58.1	56.9	55.6	66.2	97.6	43.5
$S5$	47.5	52.9	49.8	57.5	57.4	55.3	66.0	96.6	39.2
$S6$	48.1	51.3	50.6	57.9	57.4	56.5	72.7	99.8	59.4
$S7$	52.3	58.2	51.5	68.9	64.0	65.3	70.9	96.0	60.3
Overall	48.2	53.1	51.3	61.5	59.3	58.3	**72.1**	$\sim$98	$\sim$53

B. Contrastive representation

The linear probe on the frozen SupCon encoder raises accuracy to $\mathbf{53.1}\%$ (validation 58.3%): $+4.9$ points and an improvement on *every* set, the largest on $S3$ ($+8.3$, unseen person) and $S7$ ($+5.9$). Its confusion matrix (Fig. 2 (b)) sharpens *running* and softens $L \to E$ ($0.76 \to 0.52$). *Fine-tuning* then *hurts*, dropping to 48.4%: freezing the encoder is what generalizes, fine-tuning re-overfits the one environment.

The t-SNE projection (Fig. 3, from the sliding-window run of Section VI-D) is the most informative diagnostic. Colored by *environment* (right) the points are thoroughly mixed—the SupCon objective *did* reach its design goal of an environment-invariant representation. Colored by *activity* (left) the space splits into a static region (*empty*, *sitting*) and a moving arc (*jumping*, *running*, *walking*), but the activities *within* each region stay partly entangled: contrastive learning removed the *domain* nuisance and exposed a harder, intrinsic limit—a single-antenna ≈ 2 s window does not fully separate the activities, whichever way it is encoded. Consistently, in both runs the SupCon loss fell only from ≈ 4.6 to ≈ 4.3: the pre-training never fully converged.

C. Optimizer schedule

The slow schedule changes only the learning rate, patience and budget, so the two runs isolate that factor. Slowing training down *improves* accuracy from 48.2% to 51.3% ($+3.1$ points, five sets out of six) even though the model fits its own training data *worse* (0.575 vs. 0.601 training accuracy): with one training environment the two objectives conflict, as an epoch spent fitting $S1$ better is an epoch spent memorizing its room, person and day. It also tempers the previous result—$+3.1$ of the contrastive $+4.9$ points is buyable with an optimizer setting alone.

D. Sliding-window front-end

Cutting the same recordings at stride 30 instead of 340 turns $13{,}564$ windows into $152{,}320$ and takes the *unchanged* architecture from 48.2% to $\mathbf{59.3}\%$ ($+11.1$ points, $+8.4$ to $+15.3$ per set). Two caveats: the run also raises patience from 7 to 12, so part of the gain belongs to the longer schedule, and the extra windows are correlated, so the effective sample size grows far less than tenfold. The cost is ≈ 21 GB resident and an epoch an order of magnitude slower—exactly why the loader had to allocate once instead of concatenating—but the denser windows buy real fitting capacity, not noise: 0.756 training accuracy with validation 12 points above the non-overlapping run.

E. Multi-antenna decision fusion

Averaging the four per-antenna probability vectors (Eq. (2)) is the largest single improvement we measured, and it is free: no retraining, no extra parameters, one softmax and one group-average at inference. On the sliding-window model it lifts 59.3% to $\mathbf{72.1}\%$ ($+12.8$ points, from $+6.9$ on $S7$ to $+17.1$ on $S2$); on the non-overlapping model it gave $+10.2$ ($51.3 \to 61.5\%$)—robust to the data regime.

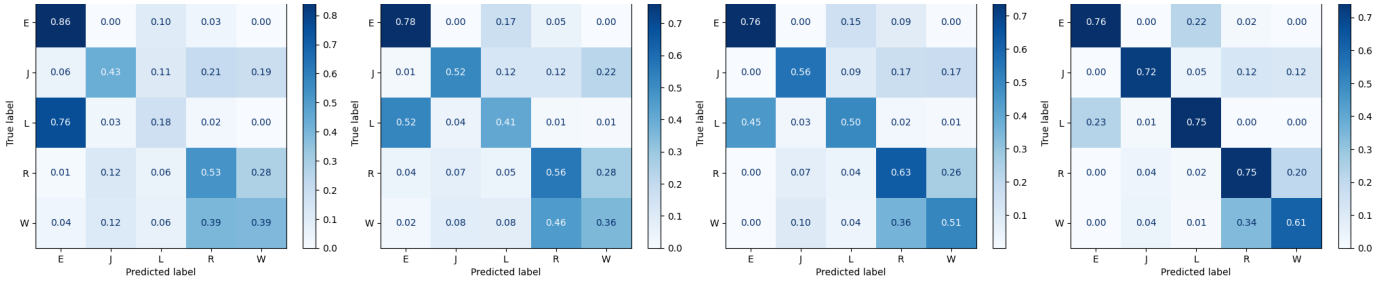
Its mechanism changes with the regime. At stride 340 fusion sharpened the motion classes but left *sitting→empty* untouched: with almost no motion the four antennas agreed on the same wrong answer. At stride 30 the single-antenna model is already half-right on that pair (Fig. 2 (c), L at 0.50) and fusion turns it into the largest per-class gain—L reaches 0.75, $L \to E$ falls from 0.45 to 0.23—next to *jumping* ($0.56 \to 0.72$) and *running* ($0.63 \to 0.75$). Averaging helps only where antennas disagree, and denser coverage is what creates that disagreement on low-motion windows.

F. Does contrastive learning still help?

Repeating the contrastive stage on the sliding-window data answers a question the first experiment could not: the probe now reaches 58.3%, 1.0 point *below* the plain baseline it had beaten by 4.9, and fine-tuning again does worse (57.0%). The stage thus behaves as a *data-efficiency* mechanism more than a domain-invariance one: with 2767 training windows it buys a representation cross-entropy cannot learn; with $31{,}103$ correlated windows cross-entropy catches up alone. One confound: its learning rate was left at 10^{-4} there while the baseline used 10^{-3}, so part of the deficit may be an under-trained encoder rather than the data regime.

G. Comparison with the SHARP paper

Tab. 2 places our results next to SHARP [1]: the *full* system (strongest-path sanitization + 4-antenna fusion) and an *ablation* that swaps the sanitization for the standard reference-antenna method, feeding the *same* CNN ([1], Table 4). Full SHARP stays above 95% everywhere. Our best configuration (72.1%) beats the ablation on *every* unseen set, by 10.6 ($S7$) to 26.8 points ($S5$), with the largest margins exactly where it



(a) baseline, non-overlapping (b) + SupCon linear probe (c) baseline, sliding window (d) + 4-antenna fusion

Fig. 2: Row-normalized confusion matrices on all unseen test environments, along the four milestones. The contrastive representation (b) softens the $L \to E$ confusion of the baseline (a); the sliding window (c) lifts every class; fusion (d) repairs the low-motion pair itself, L going from 0.50 to 0.75 with the *same* weights.

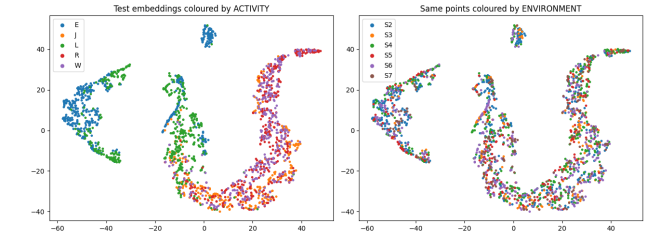


Fig. 3: t-SNE of the SupCon test embeddings (sliding-window run). *Left*, by activity: a static region (*empty*, *sitting*) and a moving arc (*jumping*, *running*, *walking*). *Right*, the same points by environment: colors are mixed *inside* the structure, i.e. the space is organized by *activity*, not by *environment*.

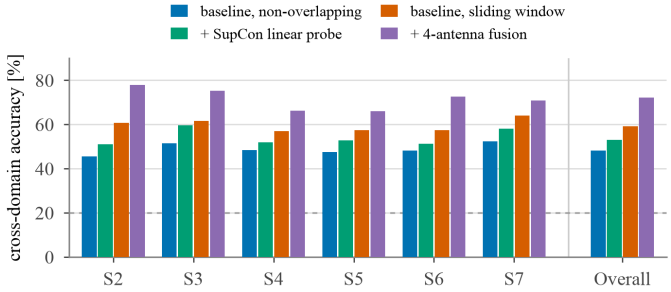


Fig. 4: Cross-domain accuracy per measurement set along the four milestones (dashed line: the 20% chance level). Every unseen set benefits from the sliding window and, on top of it, from antenna fusion.

collapses—the blocked-line-of-sight sets. The residual ~ 26-point gap to full SHARP is thus concentrated in the one ingredient we do not implement.

H. Discussion

a) What each change bought: Ranked by measured effect, decision fusion (+12.8) and the sliding window (+11.1) dominate the contrastive representation (+4.9, −1.0 once data is plentiful) and the optimizer schedule (+3.1). That ordering is a finding: the two cheapest changes—a stride constant and a few lines of post-processing—beat the one that cost the most design effort, and they compose, taking the *same* network from 48.2% to 72.1%.

b) Why we are still far from SHARP: The gap is essentially one component. SHARP uses the *strongest* multipath component as phase reference, and its ablation shows the standard reference-antenna method makes the *same* CNN collapse to 39–60% (Tab. 2, last column). Our best model sits 10 to 27 points *above* that ablation: we reproduced the decision-level half of SHARP's robustness and improved the data side, but none of its signal-processing half. Training on one small environment also makes every number high-variance.

c) Where representation learning fits: The SupCon stage attacks domain shift on the *representation* axis and, as the t-SNE shows, succeeds—but its advantage evaporates once the front-end supplies ten times more windows. It is therefore worth its complexity when labelled windows are scarce, the realistic setting for a deployment with one calibration environment; in the data-rich regime the effort is better spent on the front-end and the decision rule. Left undone: fusion on top of the frozen probe with the contrastive stage re-tuned at 10^{-3}.

VII. Concluding Remarks

Starting from a baseline Wi-Fi HAR pipeline under a strict cross-domain protocol (48.2% on unseen environments), we implemented and measured four strategies one at a time: a supervised-contrastive representation (+4.9 points, with a t-SNE confirming environment-invariance), a slower optimizer schedule (+3.1), an overlapping sliding window (+11.1) and multi-antenna soft decision fusion (+12.8). The last two compose and take the *same* 128,535-parameter network to **72.1**%, above the single-component ablation of the reference system [1] on every unseen set.

Two of them cost almost nothing at training time—a stride constant and a probability average over receiver diversity the dataset already contains—which makes them the first things to try in a new deployment. The contrastive encoder is conditional: it helps when labelled windows are scarce and stops helping once the front-end supplies ten times more, so we read it as a data-efficiency mechanism rather than a substitute for a good front-end. Still missing: the strongest-path sanitization behind most of the remaining gap, multi-seed runs, and fusion on top of the frozen probe.

a) What we learned and difficulties encountered: The strict environment-based split changes the story: modest-looking numbers are the honest cost of never seeing the test environment, which a random split would hide. Three results were counter-intuitive—training *less* well generalized *better*, fine-tuning the contrastive encoder *hurt*, and the largest gains came from the two cheapest changes once the pipeline carried

a group identifier. The main difficulties were getting SupCon pre-training to converge (the loss barely moved), the run-to-run variance of a small model trained on one environment, and the memory cost of overlapping windows, which forced a pre-allocating loader before the 91%-overlap front-end could run.

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